12
Chapter

PSYCH 301
Cognitive Processes

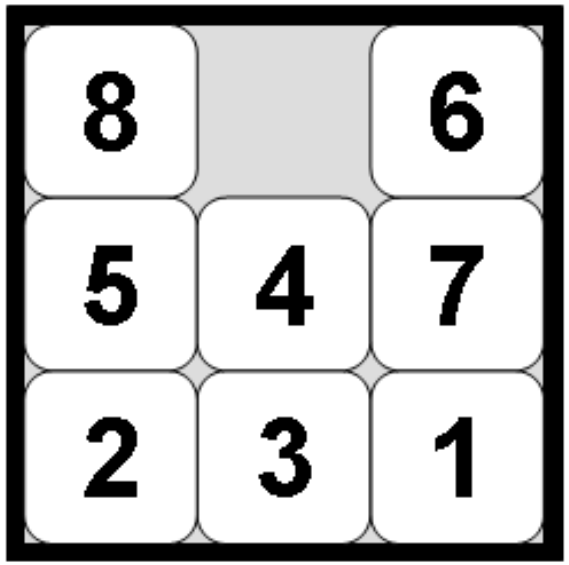
Lecture 13:
Problem-Solving and Reasoning

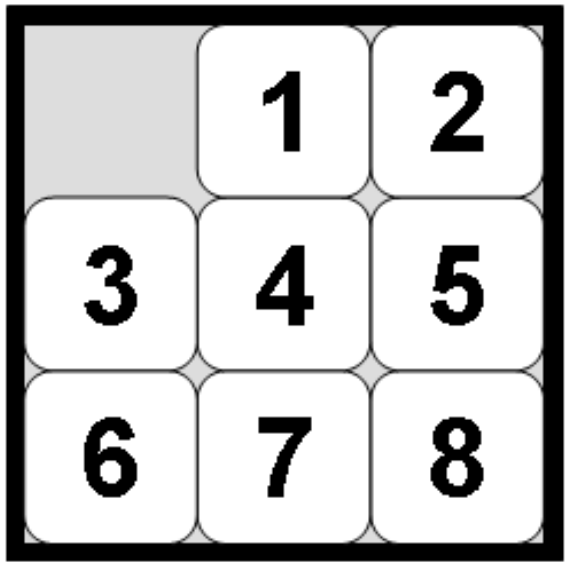
Fall 2026

Instructor: Bryan Shilowich

What Is a Problem?

- Problem:
  - An obstacle between a present state and a goal
  - Solution not immediately obvious



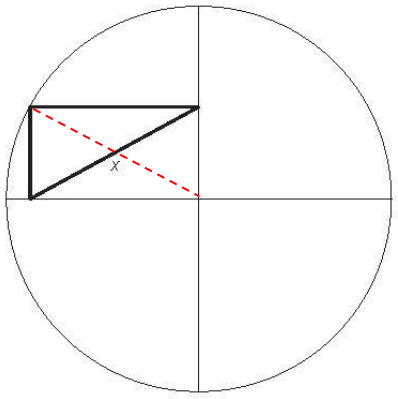


Gestalt Approach

- Gestalt psychologists were interested not only in perception but also in learning, problem solving, and even attitudes and beliefs (Koffka, 1935)
  - They took a perceptual approach to dealing with all the above

- Problem solving for Gestalt psychologists was about:
  - Representing a problem in the mind
  - Restructuring: reorganization or restructuring the problem's representation
    - Kohler's "circle" problem

- If the length of the circle's radius is r, what is the length of line x?



- Different people might represent this problem differently
- People's representations is likely to be different from the picture on this slide

Insight in Problem-Solving

- Often solutions are obtained by first perceiving the object and then representing it in a different way
  - ➔ restructuring of the problem's representation
- Gestalt psychologists related restructuring with *insight*
  - Sudden realization of a problem's solution
  - Discovering a crucial element
  - They thought solutions come to people all of a sudden
  - "Aha" experience
- Other researchers don't necessarily agree with this
  - A lot of our problem solving does not require "Aha" experiences

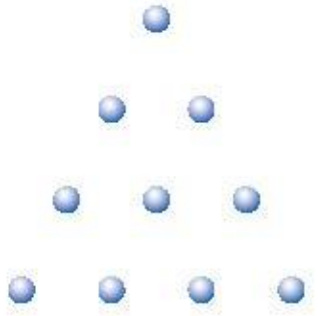
Insight in Problem-Solving

- Metcalfe and Wiebe (1987): designed an experiment to distinguish between insight and noninsight problems
  - Insight: triangle problem, chain problem
  - Noninsight: algebra
  - Hypothesis:
    - In insight problems: S should not be very good at predicting how near they are to a solution
    - In non-insight problems: involves more methodical process ➔ S should know how close they are to the solution
  - Warmth judgments every 15 seconds: “hot” ratings indicated that they are getting close to the problem

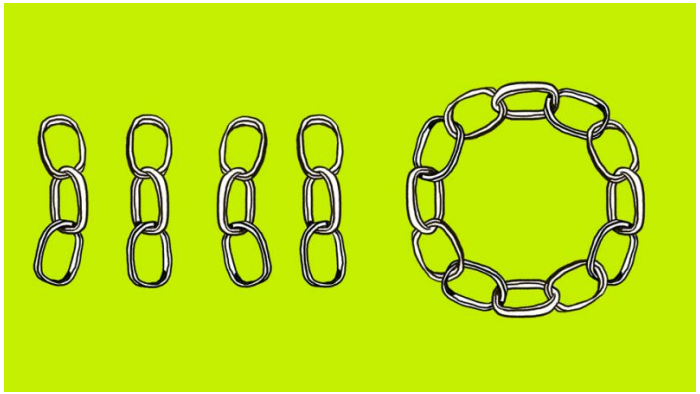
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Insight in Problem-Solving

Show how you can move three of the blue dots to get the triangle to point to the bottom of the page



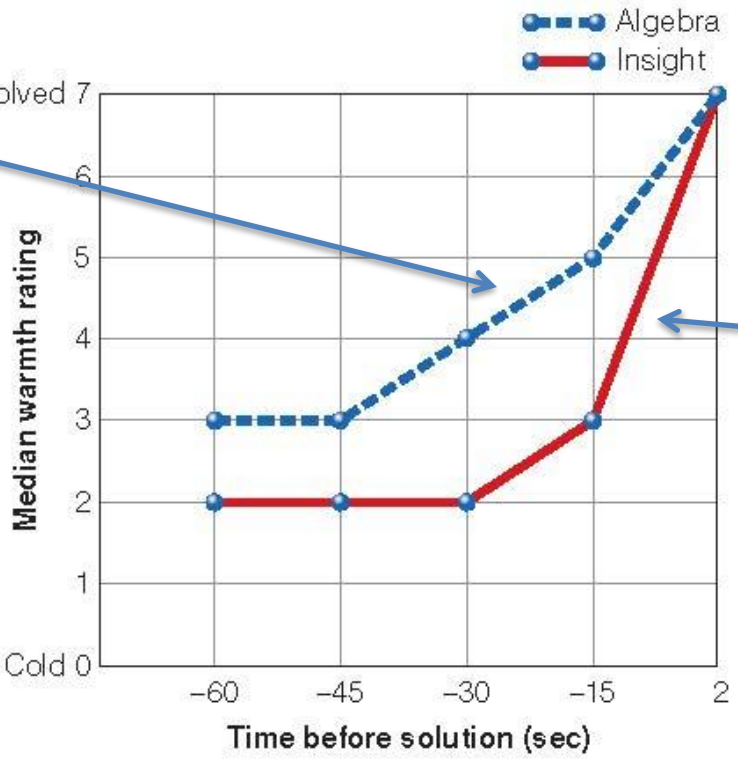
Join the pieces on the left into a single closed loop of chain (as on the right). To open a link costs 2 cents and to close a link costs 3 cents. You only have 15 cents…



Insight in Problem-Solving

- Metcalfe and Wiebe (1987) “warmer/colder” rating
  - Insight problems solved suddenly
  - Noninsight problems solved gradually

Gradual increase



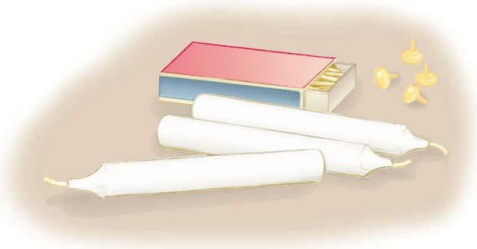
Big jump from 3 to 7

Obstacles to Problem-Solving

- Gestalt psychologists claimed that one of the major obstacles to problem solving is *fixation*
  - Tendency to focus on a specific characteristic of the problem that keeps people from arriving at a solution

- Functional fixedness: restricting use of an object to its familiar functions

Candle problem: You are in a room with a vertical corkboard mounted on the wall. You are given the material on the right. Mount a candle on the corkboard so it will burn without dripping was on the floor



Seeing matchboxes as containers inhibited using them as supports

- S who were given empty matchboxes instead, were twice as likely to solve the problem

Obstacles to Problem-Solving

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- Functional fixedness: restricting use of an object to its familiar functions

Two-string problem: Tie together the two strings

function of pliers gets in the way of seeing them as a weight
They can be used to create a pendulum



Obstacles to Problem-Solving

- Two string problem: once a hint was provided by "accidentally" setting the string in motion, most of the S were able to solve it
  - In Gestalt terms: the solution occurred when S restructures their representation

- Mental set
  - A preconceived notion about how to approach a problem
  - Based on a person's past experiences with the problem (or similar problems)
  - Water-jug problem: figure out how to obtain a required volume of water, given three empty jars for measure

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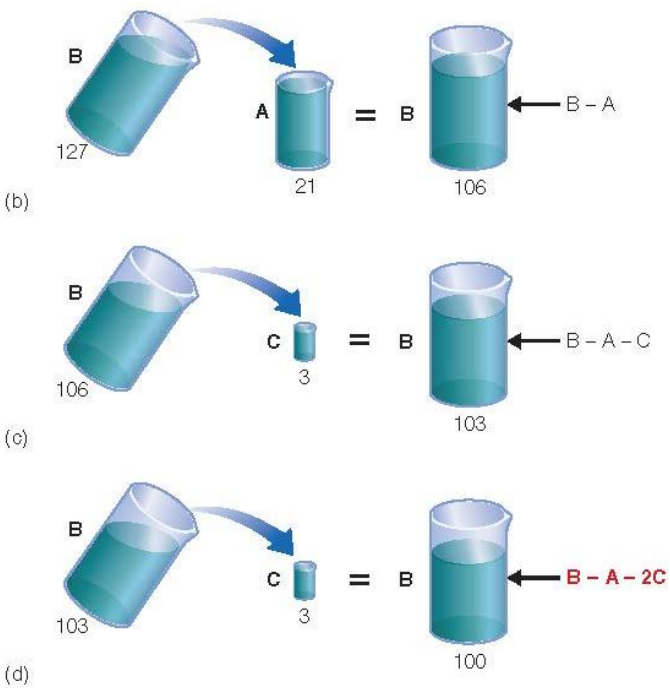
Obstacles to Problem-Solving

Each problem specifies the capacities of jugs A, B, and C and a final desired quantity.

The task is to figure out how to use the jugs with these capacities to measure out the desired quantity.

Capacities (quarts)

Problem	Jug A	Jug B	Jug C	Desired quantity
1	21	127	3	100
2	14	163	25	99
3	18	43	10	5
4	9	42	6	21
5	20	59	4	31
6	20	50	3	24
7	15	39	3	18
8	28	59	3	25



Given mental set inhibited participants from using simpler solution

Information-Processing Approach

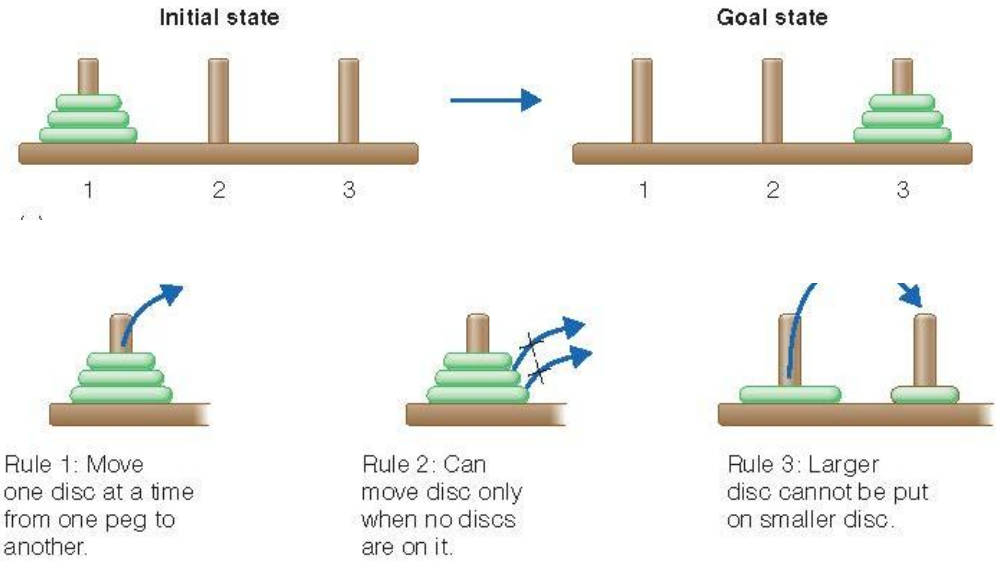
- Newell and Simon (1956): Logic theorist
- Described problem solving as a process that involves search
  - Instead of just considering the initial structure and then the new structure achieved when the problem is solved

- ➔ Newell and Simon described problem solving as a search that occurs between the posing of the problem and its solution
  - "searching for a way to reach a goal"
  - "getting around road-blocks"

- Problem space
  - Initial state: conditions at the beginning of the problem
  - Intermediate state(s)
  - Goal state: solution of the problem

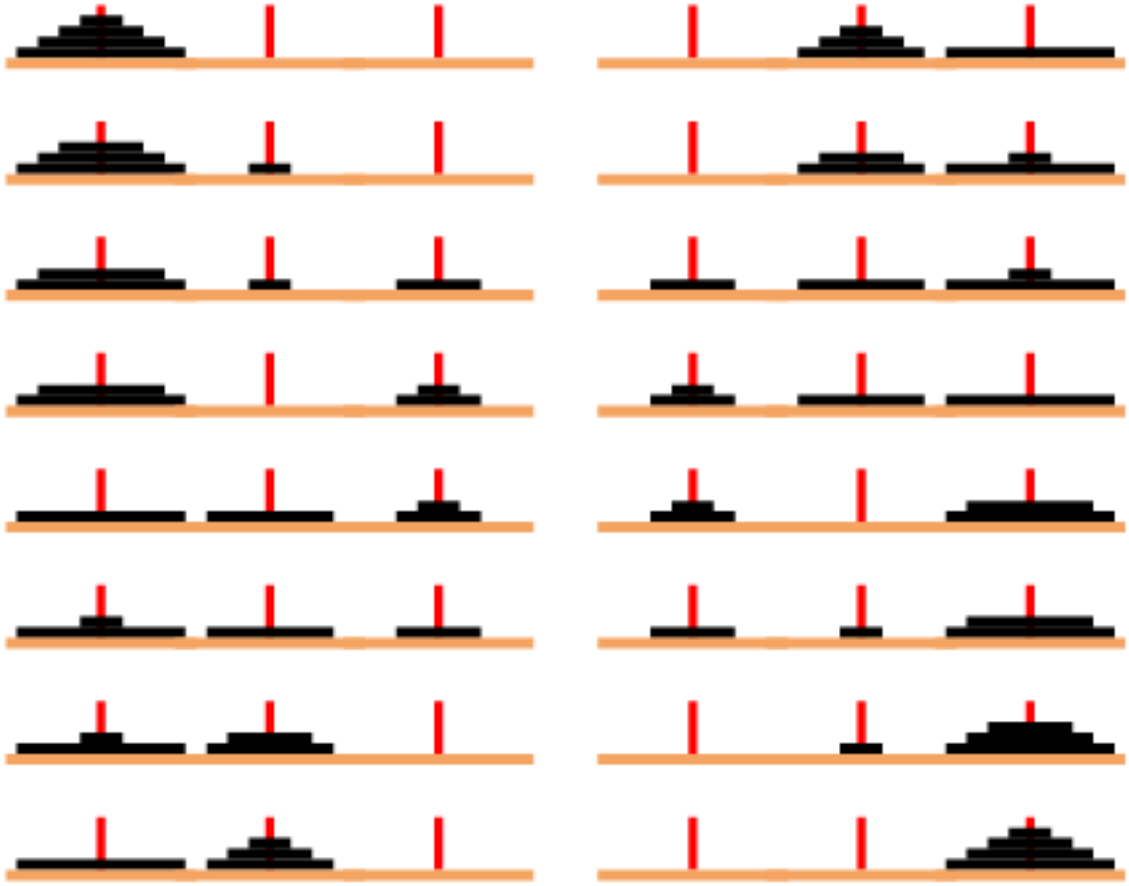
Information-Processing Approach

- Operators: rules specify which moves are allowed and which are not
  - Take the state of the problem from one state to another
- Tower of Hanoi



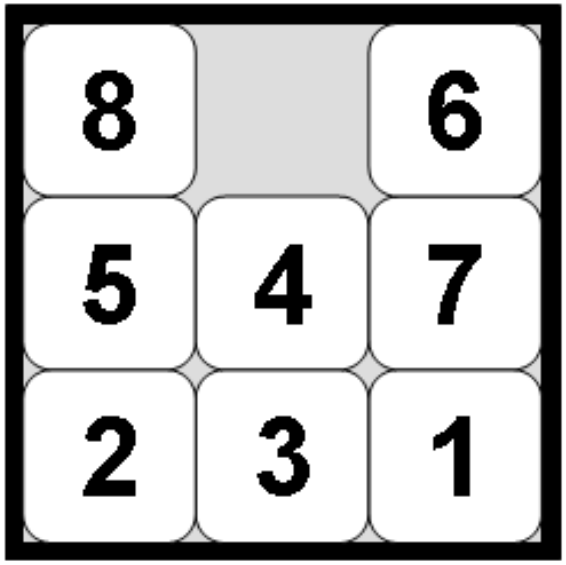
- Disks are used one at a time
- A disk can be moved only when there are no discs on top of it
- A larger disk cannot be placed on top of a smaller disk

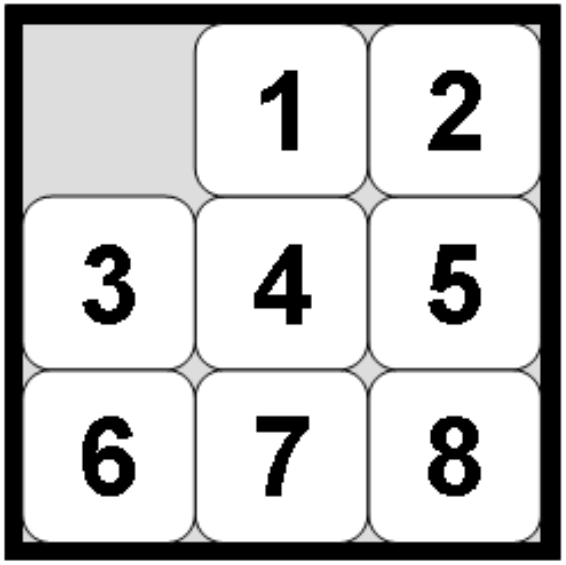
Information-Processing Approach



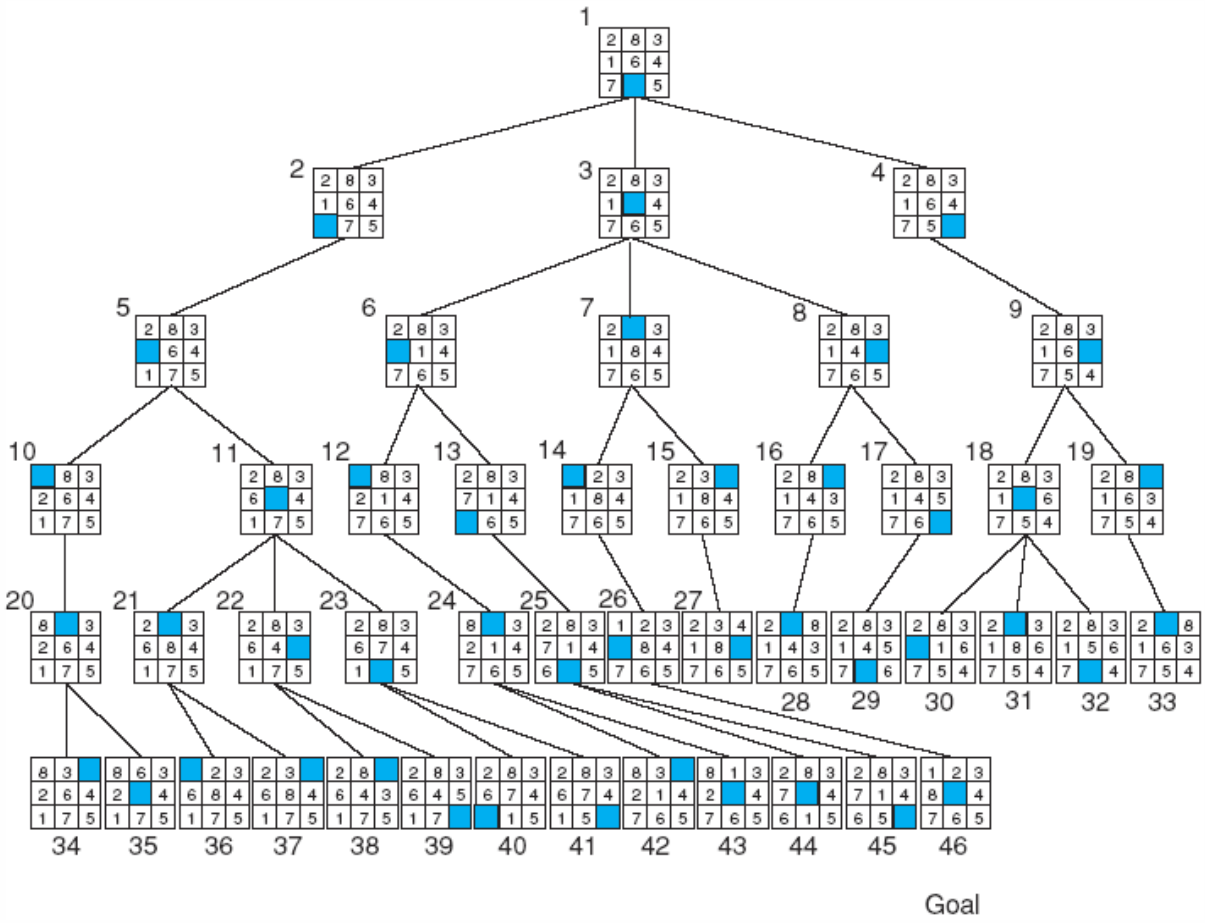
With 64 disks, it will take a trillion years to solve it, even if each moved was made in a second, and it was correct

Information-Processing Approach





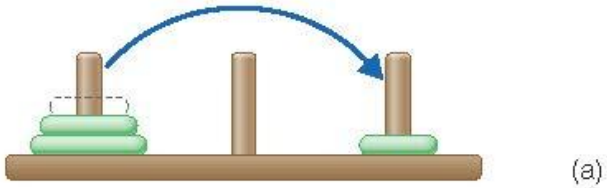
Information-Processing Approach



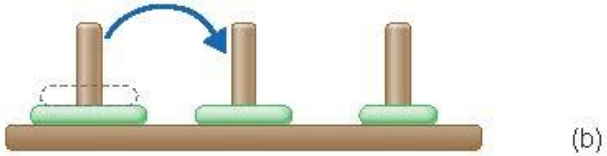
Information-Processing Approach

- According to Newell and Simon, we have to search the problem space to find a solution
- Means-end analysis: reduce differences between initial and goal states
  - Subgoals: create intermediate states closer to goal
- For Tower of Hanoi, applying means-end analysis reduces the size of the difference between initial and goal states

Information-Processing Approach

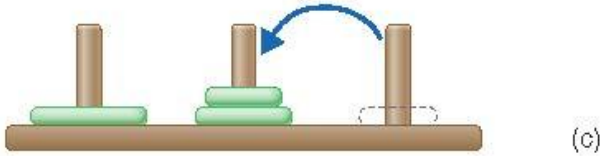


(a)



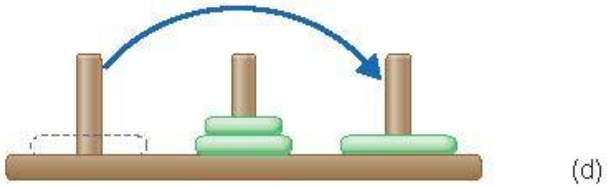
(b)

Subgoal 1: Free up large disc.



(c)

Subgoal 2: Free up third peg.



(d)

Subgoal 3: Move large disc onto third peg.

The Importance of How a Problem Is Stated

- One of the main contributions of Newell and Simon's approach: provided a way to specify the possible pathways from the initial to goal states

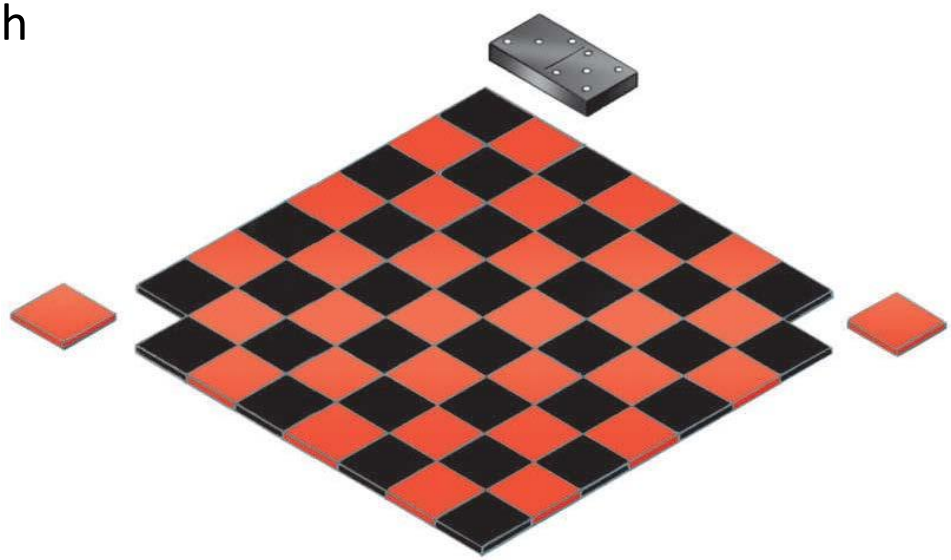
- There is more to problem solving than specifying the problem state
- How a problem is stated can affect its difficulty

- Mutilated-checkerboard problem

The Importance of How a Problem Is Stated

Checkerboard: 64 squares, can be completely covered by placing 32 dominos on the board so that each domino covers 2 squares

Mutilated checkerboard problem:
If we eliminate 2 corners, can we now cover the remaining squares with 31 dominos?



The Importance of How a Problem Is Stated

- Mutilated-checkerboard problem
  - Key to solving it: understanding that
    1. Each domino covers two squares
    2. These squares must be of different colors
  - Conditions differed in how much information provided about the squares
    1. Blank board
    2. Colored board
    3. Words *black* and *pink* on the board
    4. Words *bread* and *butter* on the board
  - Easier to solve when information is provided that points toward the correct representation of the problem
    - "Bread and butter" condition emphasized the difference between the squares
    - S solved this condition twice as fast as the blank condition

The Importance of How a Problem Is Stated

- Think-aloud protocol
  - Say aloud what one is thinking when solving a problem
    - Instructed not to describe what you are doing, but verbalize new thoughts
    - Goal: what information S is attending to while solving the problem
  - Shift in how one perceives elements of a problem
    - S get the problem after realizing that the word “bread” and “butter” are important
    - Similar to the gestalt idea of representational shift